



Statistical Sample Generator and Analyzer

This project:

1. Aims to clarify Large Number Law by generating and analyzing statistical samples from normal distributions
2. Provides tools for generating and analyzing data, including visualization and regression analysis
3. Accepts parameters for the population size, the number of samples, the number of graphs to generate, the degree of the polynomial to use for regression, and the folders to save output CSV files and graphs in



GITHUB

https://github.com/ayezza/generate_samples



Features

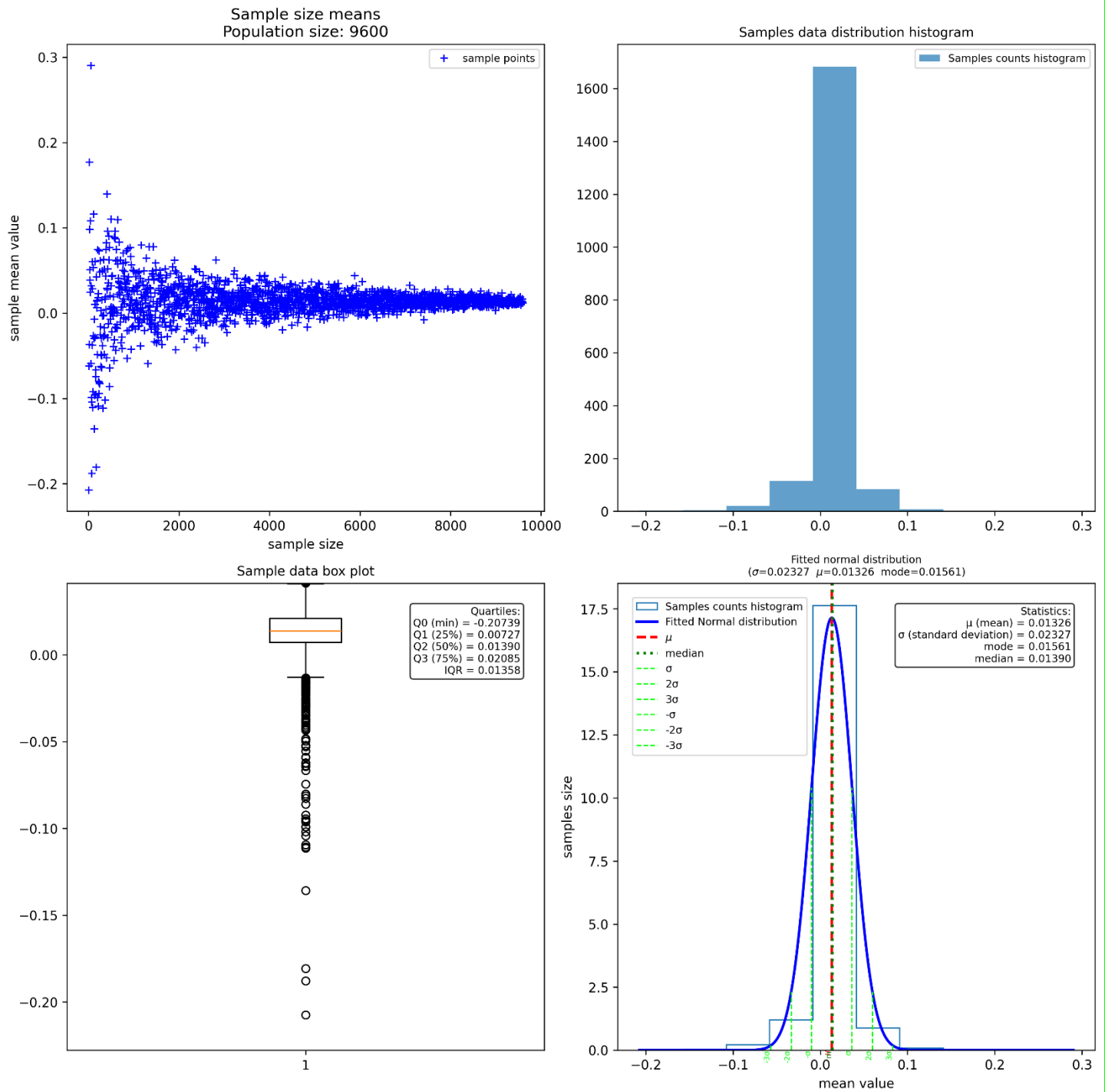
1. Generate random population samples with configurable sizes
2. Analyze sample distributions with various statistical metrics
3. Generate visualizations including:

a. Individual population analysis (4 graphs per population)

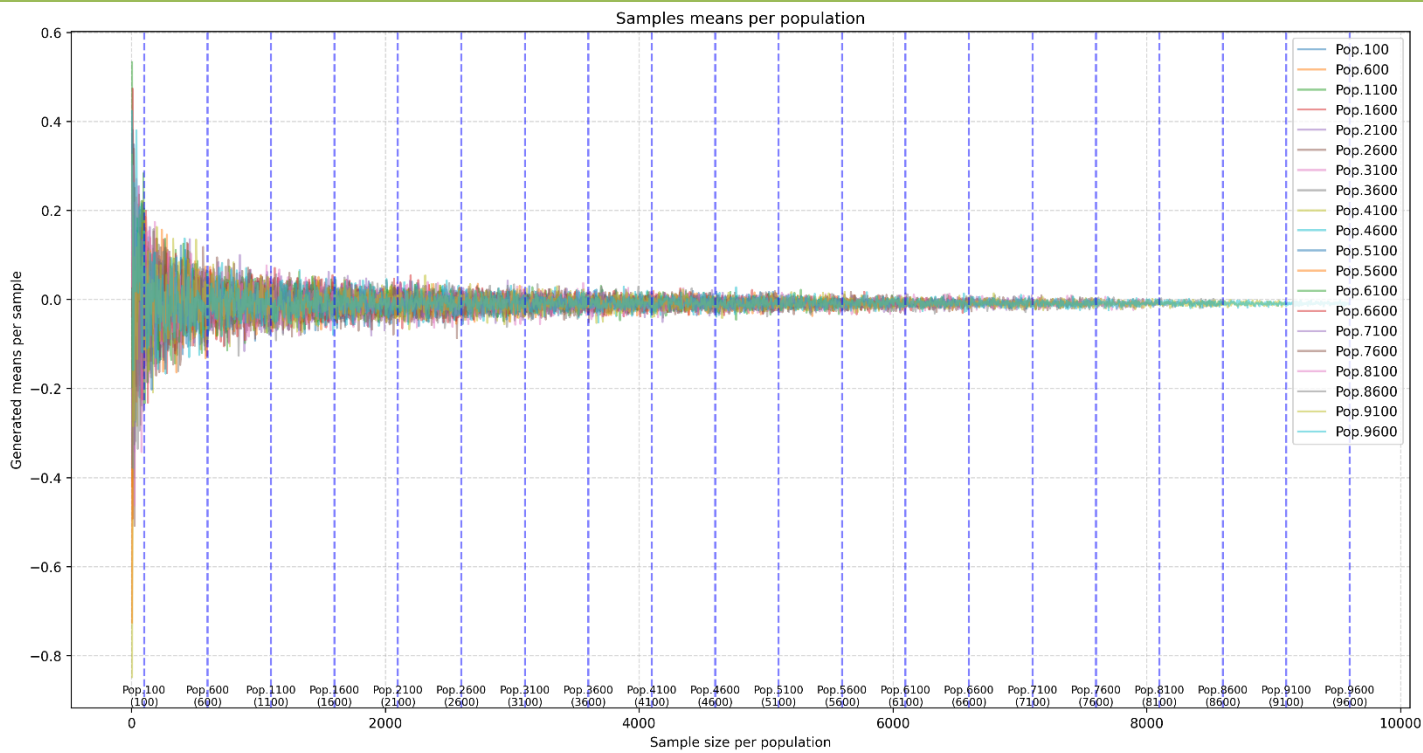
i. Sample distribution histograms

ii. Box plots

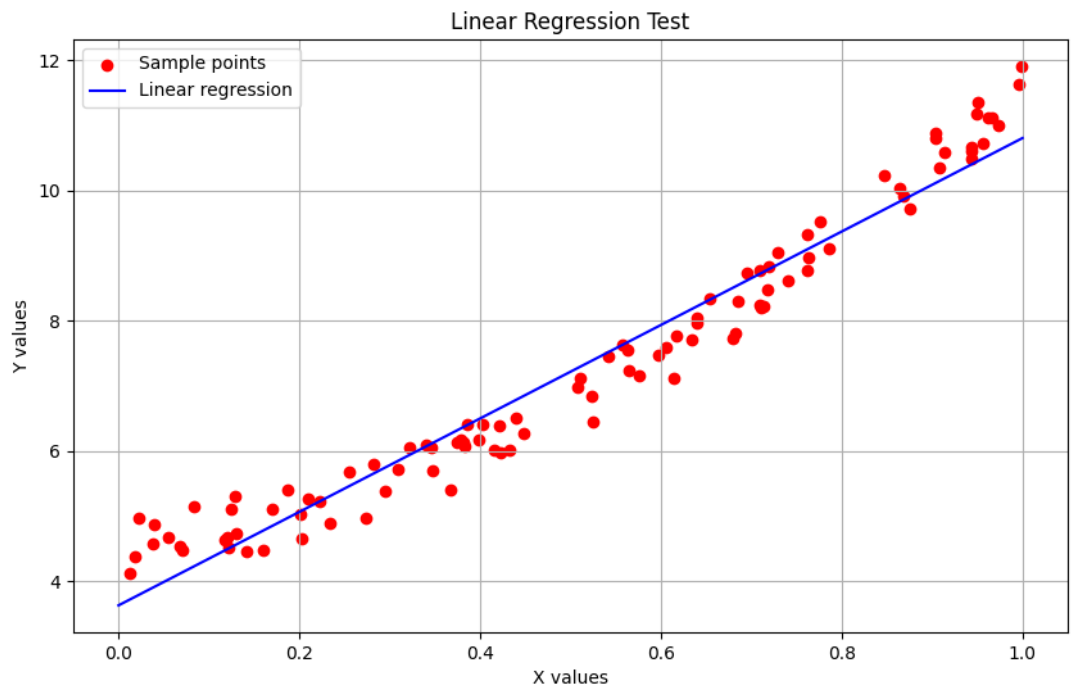
iii. Normal distribution fitting

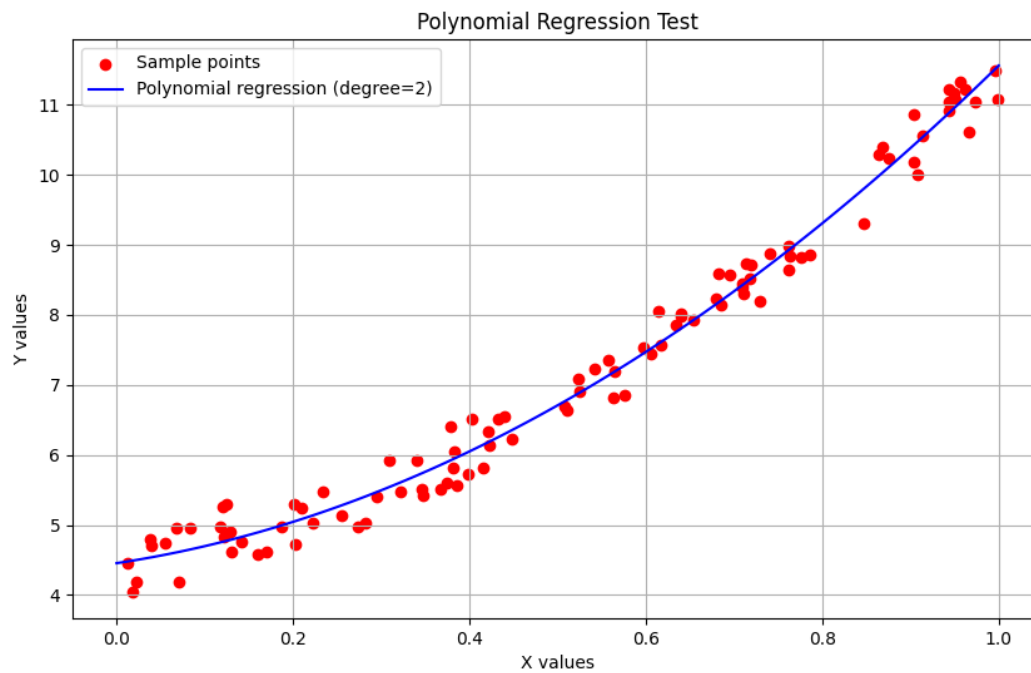


b. Combined population analysis



c- Linear and polynomial regression





4. Export results to CSV files
5. Statistical analysis including:
 - a. Mean, median, mode
 - b. Standard deviation
 - c. Distribution parameters
 - d. Confidence intervals

Installation

1. **Create a virtual environment:**

```
python -m venv venv
```

2. **Activate the virtual environment:**

- **Windows:**

```
venv\Scripts\activate
```

- **Unix/macOS:**

```
source venv/bin/activate
```

3. **Install dependencies:**

```
Pip install numpy pandas scipy scikit-learn matplotlib
```

Directory Structure

```
generate_samples/
├── LICENSE
├── README.md
├── main.py          # Main script for sample generation and analysis
├── data/            # Directory for output data files
│   └── output_10000.csv
├── graphs/          # Directory for generated graphs
│   ├── combined_populations.png
│   ├── linear_regression_test.png
│   ├── polynomial_regression_test.png
│   └── population_*.png # Individual population analysis graphs
├── test_graphs/     # Directory for test-generated graphs
│   ├── linear_regression_test.png
│   ├── polynomial_regression_test.png
│   └── population_*.png
└── test_output/     # Directory for test output files
    └── test.csv
```

Usage

Basic Execution

```
python main.py [population_size] [csv_output_file] [generate_graphs] [polynomial_degree] [graphs_folder] [show_graphs] [generate_combined_graph]
```

Parameters

population_size: Size of population (default: 1000)

csv_output_file: Path to output CSV file (default: './output.csv')

generate_graphs: Generate graphs (1) or not (0) (default: 1)

polynomial_degree: Degree for polynomial regression (default: 2)

graphs_folder: Folder for saving graphs (default: './graphs/')

show_graphs: Display graphs (1) or not (0) (default: 1)

generate_combined_graph: Graph generation mode:

1: Generate separate graphs only (default)

2: Generate combined graph only

3: Generate both separate and combined graphs

Examples

1. Default parameters:

```
python main.py
```

2. Custom population with separate graphs:

```
python main.py 10000 "output_10000.csv" 1 2 "./graphs" 0 1
```

Output:

```
Passed PARAMETERS:
population_size: 10000
csv_output_file_path: ./data\output_10000.csv
generate_graphs: True
polynomial_degree: 2
graphs_folder_name: ./graphs/
show_graphs: False
generate_combined_graph: 3
```

Temps écoulé: 35.73 secondes

3. Custom population with combined graph only:

```
python main.py 10000 "output_10000.csv" 1 2 "./graphs" 0 2
```

4. Custom population with both graph types:

```
python main.py 10000 "output_10000.csv" 1 2 "./graphs" 0 3
```

5. Run tests for interactive validation:

```
python main.py --test
```

Output:

```
=== DÉBUT DES TESTS DE LA FONCTION MAIN ===
```

```
--- Test 1: Paramètres par défaut ---
```

```
Temps écoulé: 61.28 secondes
```

```
✅ Test réussi (retour: 0)
```

✓ Graphiques générés avec succès

--- Test 2: Grande population ---

Temps écoulé: 70.33 secondes

✓ Test réussi (retour: 0)

✓ Graphiques générés avec succès

--- Test 3: Sans génération de graphiques ---

Temps écoulé: 0.08 secondes

✓ Test réussi (retour: 0)

--- Test 4: Dossier de sortie personnalisé ---

Temps écoulé: 7.87 secondes

✓ Test réussi (retour: 0)

✓ Fichier CSV généré avec succès

✓ Graphiques générés avec succès

=== FIN DES TESTS ===

Generated Files

1. CSV file containing:
 - Sample sizes
 - Population statistics
 - Mean values
 - Other statistical metrics
2. Graphs (if enabled):
 - Individual population analysis (sample_size_XXX.png)
 - Combined population analysis (combined_populations.png)
 - Normal distribution fits
 - Box plots
 - Regression analysis plots

Visualization Examples

- The program generates several types of plots depending on the generate_combined_graph parameter:
- Individual population analysis (4 graphs per population)
- Distribution analysis with μ , σ , and mode indicators
- Box plots showing quartiles and outliers
- Histograms and normal distribution fits
- Combined population analysis showing all populations in one graph
- Regression analysis for sample means

Testing

Run the automated test suite:

```
python main.py --test
```

This will execute various test scenarios and verify:

- Basic functionality
- Large population handling
- Graph generation options
- File output handling

License

This project is open source and available under the MIT License.

Contributing

Contributions are welcome! Please feel free to submit a Pull Request.

Author

Abdel YEZZA, Ph.D

License

This code is completely free and can be modified without any restriction.